

NLTK: Natural Language Processing Made Easy

We focus on the basics of natural language processing and its applications using one of the most popular NLP libraries known as Natural Language Toolkit (NLTK).



In the age of generative AI, natural language processing (NLP) is a popular subject of interest. One of the most popular NLP libraries is NLTK or Natural Language Toolkit and we will learn how to use it effectively in this article.

We will start by looking at the installation process of this library using Google Colab (<https://colab.research.google.com/>) as the IDE. Let's open the link as shown in Figure 1.

Now open a new notebook and, in the first cell, give the command `pip install nltk` to install NLTK.

Tokenisation

The first NLP process we will be looking at is tokenisation. Tokenisation is a natural language processing step in which we break a given piece of text into smaller parts. These parts can be simple phrases, sentences, words, and even characters or sets of characters, and are called tokens. Tokenisation is an important preprocessing step in most NLP applications. This may be done for feature engineering, text preprocessing or to build vocabulary for tasks like sentiment analysis.

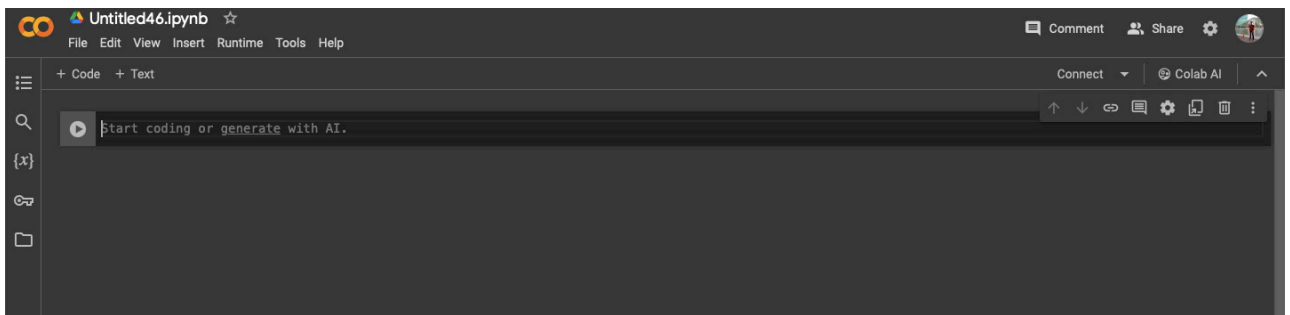


Figure 1: Google Colab